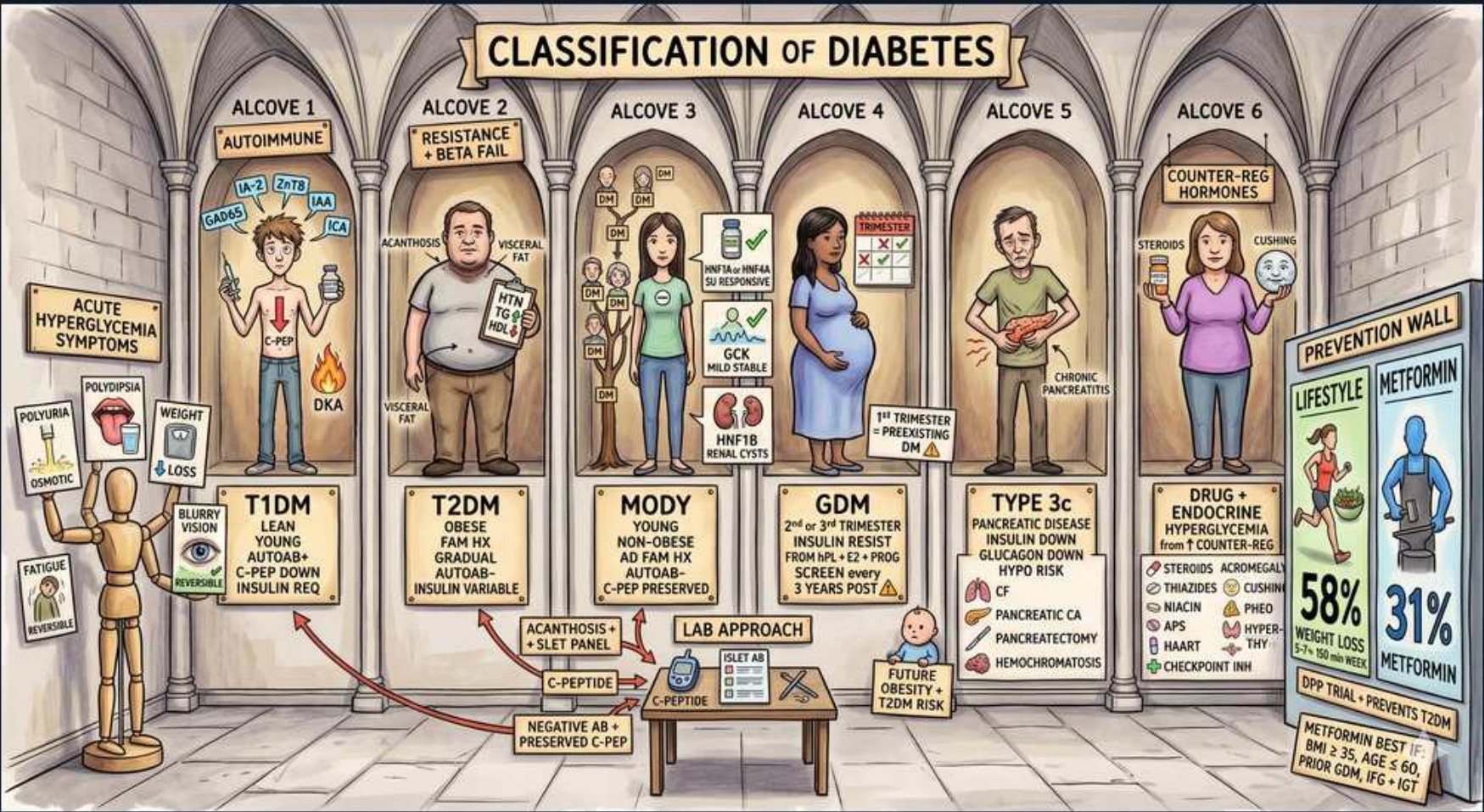


Diabetes — Scene 3: Classification Hall (Pathophys & Dx)



1. Alcove 1 — T1DM

WHAT YOU SEE

Alcove 1 'Autoimmune': GAD/IA-2/ZnT8 antibodies, C-peptide down, DKA; lean young, insulin required.

WHAT IT ENCODES

T1DM = autoimmune T-cell-mediated destruction of pancreatic beta cells, typically lean and young (but any age). Markers: GAD65, IA-2 (ICA512), insulin (IAA), and ZnT8 autoantibodies; LOW/undetectable C-peptide; DKA-prone; absolute insulin dependence. Associated with HLA-DR3/DR4 and other autoimmune disease (Hashimoto, celiac, Addison).

BOARD TRAP

WRONG: 'A lean 19-year-old with new hyperglycemia, weight loss, and ketones has T2DM because A1c is only 8%.' Lean, ketotic, antibody-positive presentation is T1DM regardless of A1c level. Discriminator: positive islet autoantibodies + low C-peptide + DKA-proneness define T1DM, not the A1c value.

2. Alcove 2 — T2DM

WHAT YOU SEE

Alcove 2 'Resistance + beta fail': obese, acanthosis, visceral fat, gradual; autoantibodies negative, C-peptide preserved.

WHAT IT ENCODES

T2DM = peripheral INSULIN RESISTANCE plus progressive beta-cell secretory failure. Risk: obesity (especially visceral fat), metabolic syndrome, acanthosis nigricans, family history, age. Autoantibodies NEGATIVE; C-peptide preserved or HIGH early. Onset is gradual; HHS more typical than DKA. Accounts for ~90-95% of diabetes.

BOARD TRAP

WRONG: 'An obese adult with hyperglycemia must have T2DM and never needs antibody testing.' Adults can have LADA (autoimmune) presenting like T2DM but failing oral agents. Discriminator: T2DM = negative autoantibodies + preserved C-peptide; if a 'T2DM' patient is lean, antibody-positive, or rapidly insulin-requiring, reconsider LADA/T1DM.

3. Alcove 3 — MODY

WHAT YOU SEE

Alcove 3 'MODY': young non-obese, autoantibody negative, C-peptide preserved; HNF1A renal cysts, GCK mild stable.

WHAT IT ENCODES

MODY = Maturity-Onset Diabetes of the Young: monogenic, AUTOSOMAL DOMINANT, onset <25 yr, non-obese, autoantibody NEGATIVE, C-peptide preserved. GCK (MODY2) = mild, stable, asymptomatic fasting hyperglycemia needing no treatment. HNF1A (MODY3) = sulfonylurea-responsive, glycosuria. HNF1B (MODY5) = renal cysts/GU anomalies.

BOARD TRAP

WRONG: 'A lean 16-year-old with mild fasting hyperglycemia, negative antibodies, and an affected parent and grandparent should start insulin for presumed T1DM.' This multigenerational, antibody-negative, non-ketotic pattern is MODY. Discriminator: AD family history across generations + negative antibodies + preserved C-peptide = MODY (GCK needs no Rx; HNF1A responds to sulfonylureas).

4. Alcove 4 — GDM

WHAT YOU SEE

Alcove 4 'GDM': 2nd-3rd trimester insulin resistance from hPL; screen at 24-28 wks; rescreen 6 wks postpartum.

WHAT IT ENCODES

Gestational DM = insulin resistance driven by placental hormones (human placental lactogen, progesterone, cortisol) in the 2nd/3rd trimester. Screen ALL pregnant women at 24-28 weeks (1-step 75g OGTT or 2-step 50g then 100g). Rescreen at 4-12 weeks postpartum; GDM markedly raises lifetime T2DM risk.

BOARD TRAP

WRONG: 'Hyperglycemia detected at the first prenatal visit is gestational diabetes.' First-trimester hyperglycemia reflects PREEXISTING (overt) diabetes, not GDM, because placental insulin resistance has not yet developed. Discriminator: GDM arises in the 2nd/3rd trimester; early hyperglycemia = pregestational diabetes.

5. Alcove 5 — Type 3c (pancreatic)

WHAT YOU SEE

Alcove 5 'Type 3c': pancreatic disease, insulin AND glucagon down, hypoglycemia-prone; chronic pancreatitis, pancreatic CA, hemochromatosis.

WHAT IT ENCODES

Type 3c (pancreatogenic) DM: destruction of islet mass causes loss of BOTH insulin AND glucagon (and pancreatic polypeptide), producing brittle, hypoglycemia-prone diabetes plus exocrine insufficiency (steatorrhea, low fecal elastase). Causes: chronic pancreatitis, pancreatic cancer, hemochromatosis ('bronze diabetes'), cystic fibrosis, pancreatectomy.

BOARD TRAP

WRONG: 'A patient with chronic pancreatitis and new diabetes can be managed exactly like ordinary T2DM.' Loss of glucagon makes them prone to severe hypoglycemia, and they need pancreatic enzyme replacement. Discriminator: Type 3c = insulin AND glucagon deficiency (high hypoglycemia risk) + exocrine insufficiency, unlike typical T2DM.

6. Alcove 6 — Drug / endocrine

WHAT YOU SEE

Alcove 6 'Counter-reg hormones': Cushing/steroids; hyperglycemia from counter-regulatory excess.

WHAT IT ENCODES

Secondary DM from excess COUNTER-REGULATORY (anti-insulin) hormones: glucocorticoids/Cushing syndrome (cortisol), acromegaly (GH), pheochromocytoma (catecholamines), glucagonoma (glucagon, with necrolytic migratory erythema), and immune-checkpoint-inhibitor-induced diabetes (can be fulminant/DKA). Treating the underlying cause often reverses the hyperglycemia.

BOARD TRAP

WRONG: 'New diabetes with central obesity, striae, and proximal weakness is just typical T2DM.' That picture suggests Cushing syndrome causing secondary hyperglycemia. Discriminator: screen for an endocrine cause (cortisol, GH/IGF-1, metanephries) when DM presents with features of hormone excess.

7. Lab approach — C-peptide

WHAT YOU SEE

'Lab Approach' bench: C-peptide; negative AB + preserved C-pep -> table.

WHAT IT ENCODES

C-peptide (cleaved 1:1 from proinsulin) measures ENDOGENOUS insulin production and is unaffected by exogenous insulin. Preserved/high C-peptide = T2DM/MODY/early disease; low/undetectable C-peptide = T1DM or longstanding beta-cell failure. Pair with islet autoantibodies to confirm autoimmunity.

BOARD TRAP

WRONG: 'Use a serum INSULIN level to gauge native beta-cell function in a patient taking insulin injections.' Assayed insulin includes the exogenous drug; only C-peptide reflects endogenous secretion. Discriminator: C-peptide (not insulin level) distinguishes endogenous reserve in insulin-treated patients.

8. Drug list — secondary causes

WHAT YOU SEE

Steroids, thiazides, niacin, beta-blockers, statins, antipsychotics, checkpoint inhibitors.

WHAT IT ENCODES

Drugs causing/unmasking hyperglycemia: glucocorticoids (most potent), thiazides (hypokalemia impairs insulin release), niacin, atypical antipsychotics (olanzapine, clozapine), statins (mild), protease inhibitors, calcineurin inhibitors (tacrolimus), and immune checkpoint inhibitors (can cause autoimmune insulin-deficient DM/DKA).

BOARD TRAP

WRONG: 'Beta-blockers and statins should be stopped in any diabetic because they raise glucose.' Their cardiovascular benefit far outweighs the small glycemic effect, and they are indicated in diabetics. Discriminator: do not withhold guideline-indicated statins/beta-blockers over modest glucose effects; adjust diabetes therapy instead.

9. Prevention wall — lifestyle 58%

WHAT YOU SEE

Prevention wall: lifestyle 58% weight loss; metformin 31%.

WHAT IT ENCODES

Diabetes Prevention Program: intensive LIFESTYLE intervention ($\geq 7\%$ weight loss + ≥ 150 min/week activity) reduced progression from prediabetes to T2DM by $\sim 58\%$, versus $\sim 31\%$ with metformin. Lifestyle is the single most effective intervention for prediabetes.

BOARD TRAP

WRONG: 'For a patient with prediabetes (A1c 6.0%), metformin is the first and most effective intervention.' Lifestyle modification is first-line and more effective (58% vs 31%). Discriminator: lifestyle beats metformin for prevention; metformin is adjunctive for higher-risk subgroups.

10. Metformin best in subgroups

WHAT YOU SEE

Metformin best if BMI>35, age<60, prior GDM, high fasting glucose.

WHAT IT ENCODES

Metformin's prevention benefit is GREATEST in BMI ≥ 35 , age <60, women with prior GDM, and those with higher fasting glucose/A1c — in these groups it approaches lifestyle efficacy. Consider metformin for prediabetes especially when lifestyle alone fails.

BOARD TRAP

WRONG: 'A frail 70-year-old with prediabetes and BMI 24 should be started on metformin to prevent diabetes.' Metformin's preventive benefit is concentrated in younger, obese, or prior-GDM patients, not lean elderly. Discriminator: target metformin prevention to BMI ≥ 35 , age <60, or prior GDM; otherwise emphasize lifestyle.

11. Acute hyperglycemia symptoms

WHAT YOU SEE

Symptom column: polyuria, polydipsia, weight loss, blurry vision, fatigue.

WHAT IT ENCODES

Classic hyperglycemia symptoms: polyuria and polydipsia (osmotic diuresis once glucose exceeds the renal threshold ~ 180 mg/dL), unintentional weight loss, blurry vision (osmotic lens swelling), and fatigue. Diagnosis: fasting glucose ≥ 126 , 2-h OGTT ≥ 200 , A1c $\geq 6.5\%$, or random glucose ≥ 200 with symptoms (confirm unless unequivocal).

BOARD TRAP

WRONG: 'A single random glucose of 150 mg/dL in an asymptomatic patient is diagnostic of diabetes.' That value is non-diagnostic; diagnosis requires fasting ≥ 126 , A1c $\geq 6.5\%$, OGTT ≥ 200 , or random ≥ 200 WITH symptoms, with confirmation. Discriminator: an asymptomatic abnormal test must be confirmed on a second occasion.